

Heart-lung sound separation by nonnegative matrix factorization and deep learning

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ABSTRACT

Auscultation is an ordinary way to diagnose cardiovascular and respiratory diseases. The heart-lung sound (HLS) from the stethoscope is a mixture of heart sound (HS) and lung sound (LS). The first and most crucial step is to differentiate between HS and LS from the mixture signal. This paper proposes an unsupervised single-channel blind source separation (BSS) algorithm based on nonnegative matrix factorization (NMF) and deep learning (DL). Firstly, HS signals are extracted by the multi-constrained NMF (MCNMF) decomposition algorithm and K-means & SVM (KS) clustering method proposed in this paper, and different types of HS signals are classified by a convolutional neural network (CNN). Then, the HLS is separated by the embedding space centroid network (ECNet) proposed in our study. The method outperforms the known state-of-the-art techniques in all evaluation metrics SNR, SDR, SIR, SAR, and correlation coefficient. Our algorithm has two advantages: 1) this study is the first to combine NMF and DL for the HLS separation. MCNMF with selected constraints can have a more substantial sparsity effect that improves the decomposition accuracy. KS clustering algorithm can distinguish the HS component from the LS component in two clustering results. By mapping classified HLS into an embedding space, the ECNet demonstrates excellent separation results. 2) HS extraction and classification first, then HLS separation. Because our method exploits more detailed information from HS types in HLS before BSS, the ECNet can outperform other HLS separation methods.

1. Introduction

THE incidence of cardiovascular and respiratory disorders is increasing year by year, and cardiopulmonary disease has become one of the major disorders that seriously threaten human life and health [1–7]. Some cardiopulmonary diagnostic methods are invasive and may damage the human body to a certain extent. Cardiopulmonary auscultation is a non-invasive, clinically standard approach [8–12]. HS and LS are common sources of the diagnostic basis for cardiopulmonary disease [13–14], containing pathological facts about the cardiovascular and respiratory information, respectively [15]. Still, during clinical auscultation, HS and LS overlap severely in the frequency and time domains [16–17], which makes the conventional BSS unable to separate HLS effectively [18]. Meanwhile, the diversity of HS depending on the site of auscultation and the patient's lesion, and the difference in the energy distribution of various types of HS, often lead to different characteristics of HLS [19]. Therefore, the study of BSS methods applied to HLS separation has great clinical significance and is used to improve the quality

of auscultation and assist physicians in aiding diagnosis. Domestic and international scholars have studied many HLS separation methods.

Researchers have studied many BSS methods of the HLS separation [20–24]. Makkiabadi *et al.* proposed a convolutional BSS method using an orthogonal model [25]. Takada *et al.* introduced the higher-order statistics (HOS) method for HLS separation [26]. These methods are mainly based on some a priori assumptions, which are greatly restricted when applying them. Adaptive signal processing (ASP) can be used in HLS separation by introducing an optimality criterion in signal processing, thus enhancing the desired signal and eliminating the interference signal [27–28]. Nersisson *et al.* proposed an improved least mean square (LMS) algorithm based on an adaptive noise offset technique to separate HS and LS [29]. Al-Naggar *et al.* proposed an adaptive noise cancellation with a normalized last-mean-square (ANC-NLMS) algorithm to HLS BSS [30]. However, the ASP-based methods achieve HLS separation with a reference signal, which is not easy to obtain in practice. Independent component analysis (ICA) is a computational method for separating multivariate signals into additive subcomponents and is a

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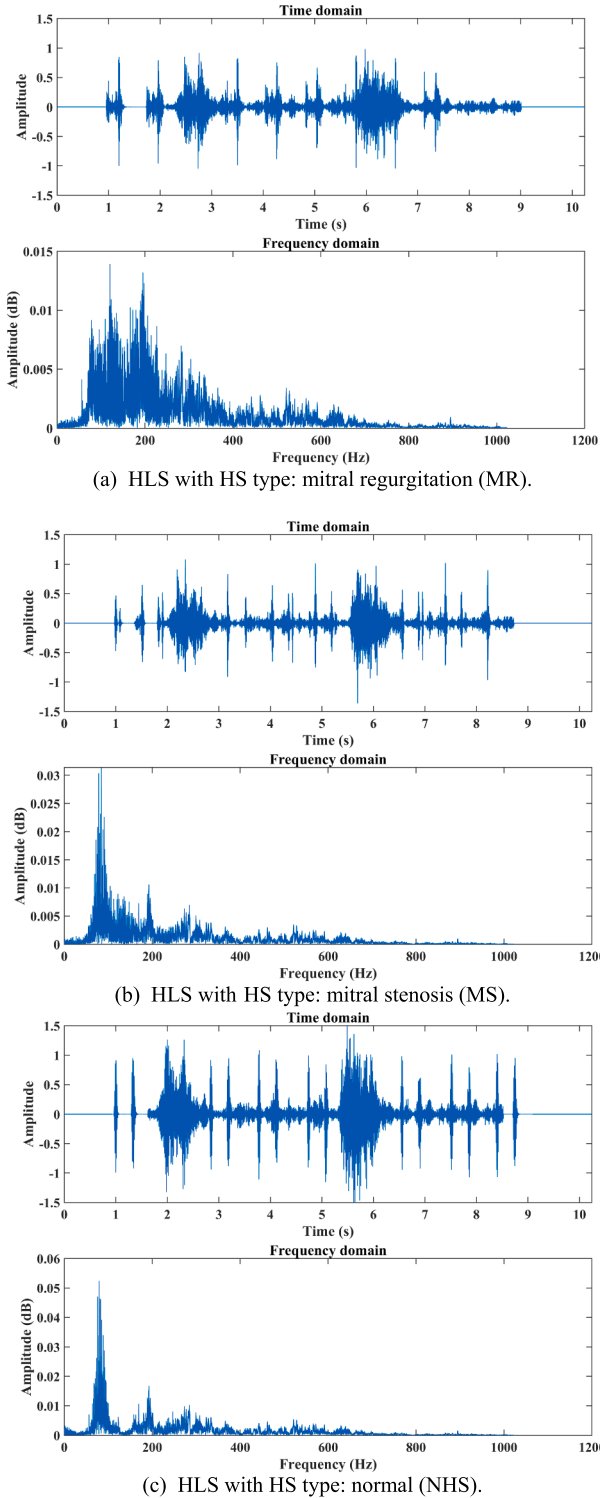


Fig. 1. HLS in the time and frequency domains with different types of HS.

commonly used BSS method for separating HLS [31–33]. However, the ICA-based methods can not apply to single-channel HLS separation. NMF is a typical method for BSS that can acquire recurring patterns from the transform domain, and its characteristic of separating signals without a priori conditions is suitable for single-channel HLS. Shah *et al.* proposed an NMF-based HLS separation method with two unsupervised clustering methods [34]. Lin *et al.* performed an NMF-based to separate HLS that processed multivariate data by constant Q transform (CQT) for dimensionality reduction [35]. Shah *et al.* introduced an improved NMF

with shared factors (NMF-SF) and an unsupervised BSS method that does not require any training data [36]. Canadas *et al.* proposed a spectral-temporal clustering by NMF to extract HS from HLS [37]. Montoro *et al.* designed a parallel source separation system based on NMF for extracting HS and LS from a single-channel signal [38].

Deep learning has been gained attention for its excellent ability to acquire a non-linear mapping between input and target. In recent years, neural networks have achieved superior results over the traditional BSS method. Sathesh *et al.* proposed a method for separating real-time HS signals from LS signals using neural networks [39]. Lei *et al.* introduced a fully connected long short-term memory (LSTM) to separate HLS [40]. Tsai *et al.* studied a periodically encoded deep autoencoder (PC-DAE) method for HLS separation in an unsupervised way by assuming different periodicity between heart and respiration rate [41]. The PC-DAE methods benefit from a DL-based model and achieve better separation results.

Although NMF methods can separate single-channel HLS in unsupervised learning and have the advantage of BSS without a priori conditions, the characteristics of the mathematical operations during matrix decomposition make NMF unable to utterly separate HLS because of the severe overlapping in the time domain and frequency domain. Meanwhile, DL methods are well suited for mining data features during HLS separation, but the diversity of HLS makes the neural network less robust.

To address the effective separation of single-channel HLS without a priori conditions and exploit more detailed information during BSS, we propose a method based on NMF and deep learning. Section 2 describes the research background. The framework and rationale of our proposed method are detailed in Section 3. Section 4 presents the experimental results. Conclusions are given in Section 5.

2. Research background

2.1. Heart and lung sounds background

HS and LS signals are mechanical vibration signals. HS is generated by the opening and closing of the heart valves. LS is caused by the airflow turbulence through the airways and alveoli. Abnormal HS can mainly be classified as aortic stenosis (AS), mitral stenosis (MS), mitral regurgitation (MR), mitral valve prolapse (MVP), etc. LS can be divided into alveolar breath (AB), bronchial breath (BB), wet rales (WR), rales (RA), etc. The HS and LS signals interfere in both the time domain and frequency domain, making HLS challenging to separate. The frequency ranges of normal and abnormal HS are 20–150 Hz and 100–1000 Hz, respectively. The frequency range of LS is 100–1000 Hz [41]. In addition, Fig. 1 shows HLS with different types of HS in the time domain and frequency domain. The different types of HS can bring about different characteristics of HLS.

2.2. NMF

NMF is a dimensionality reduction algorithm that aims to decompose a high-dimensional matrix into two lower-dimensional matrices, a matrix decomposition algorithm under the constraint that the elements are non-negative [44]. By requiring all aspects of the original matrix to be non-negative in NMF, the non-negative high-rank matrix V can be decomposed into the product of two smaller non-negative low-rank matrices W and H . This process can be expressed as.

$$V \approx WH \quad (1)$$

where $V \in R^{M \times N}$, $W \in R^{M \times K}$, $H \in R^{K \times N}$. V is the original matrix, the column vectors of V are the weighted sum of all column vectors in W , so W is called the basis matrix, and the weight coefficients are the elements of the corresponding column vectors in H , so H is called the coefficient matrix.

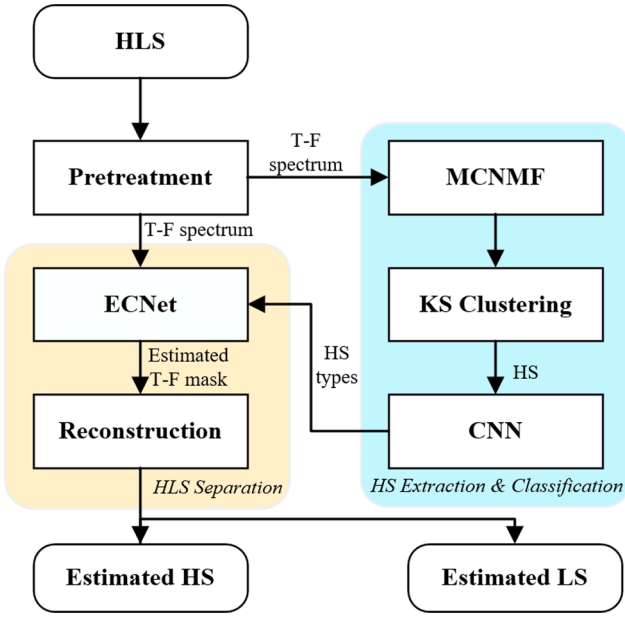


Fig. 2. Flowchart of HLS separation based on MCNMF and ECNet.

Lee and Seung proposed iterative optimization algorithms based on different distance measures for convergence of the objective function to local extremum points [45], namely the Euclidean distance-based NMF and the KL scatter-based NMF. If the errors in V and WH obey a Gaussian distribution, it is appropriate to choose an NMF based on Euclidean distance for the solution. If the errors of V and WH obey Poisson distribution, the KL scatter based NMF is generally chosen for the solution. The Itakura-Saito (IS) divergence belongs to the class of Bregman divergences and is a limit case of the β -divergence. Since the IS divergence is scale-invariant, a property well suited to speech signals with an extensive dynamic range of V components, researchers have proposed an NMF based on IS divergence (NMF-IS).

2.3. Deep learning

CNN is a particular type of feed-forward neural network that contains many convolutional computations inside the CNN. It has a deep design, making them a very mature network framework for deep learning today.

Recurrent Neural Networks (RNN) input data is time-series and time-dependent, meaning that data at one point in time impacts later data. The RNN, therefore, not only takes into account the previous moment's input but also gives the network a "memory" function of the previous content.

LSTM networks, also known as extended short-term memory networks, are a particular type of RNN mainly used to solve the gradient disappearance and gradient explosion problems in long sequence training. LSTM single unit contains three gates, input, output and forget, which are used to control the flow of information into or out of their memory. LSTM is appeared as more efficient in handling non-linearity in larger datasets. As an extension to RNNs, Bi-LSTM can be formed using LSTM combined with inverse sequences. In this recursive feedback neural network, both past and future are hidden layer states, allowing hidden connections between time-series data to be discovered.

2.4. Evaluation metrics

In this paper, there are some evaluation metrics to assess the effectiveness of HLS separation.

1. Correlation coefficient is given by

$$Cor = \frac{|\sum_i^{c,r} s_i(t) \hat{s}_i(t)|}{\sqrt{\sum_i^{c,r} s_i^2(t) \sum_i^{c,r} \hat{s}_i^2(t)}} \quad (2)$$

where subscript c, r represent HS and LS, $s_i(t)$ is the source signal, and $\hat{s}_i(t)$ indicates the estimated signal.

2. Signal to noise ratio (SNR) is defined as

$$SNR = 10 \log_{10} \left(\frac{\sum_i^{c,r} s_i^2(t)}{\sum_i^{c,r} [s_i(t) - \hat{s}_i(t)]^2} \right) \quad (3)$$

3. In the BSS, the estimated signal can be decomposed into

$$\hat{s}_i(t) = s_{target}(t) + e_{interf}(t) + e_{noise}(t) + e_{artif}(t) \quad (4)$$

where $s_{target}(t)$ is the estimate of the target source signal within the acceptable distortion range, $e_{interf}(t)$ is the estimate of the interferer signal, $e_{noise}(t)$ is the estimate of the noise signal, and $e_{artif}(t)$ is the error. Thus, three normalized BSS evaluation metrics [42], signal distortion ratio (SDR), signal to interferences ratio (SIR), and signal to artifacts ratio (SAR), can be calculated by

$$SDR = 10 \log_{10} \frac{\|s_{target}(t)\|^2}{\|e_{interf}(t) + e_{noise}(t) + e_{artif}(t)\|^2} \quad (5)$$

$$SIR = 10 \log_{10} \frac{\|s_{target}(t)\|^2}{\|e_{interf}(t)\|^2} \quad (6)$$

$$SAR = 10 \log_{10} \frac{\|s_{target}(t) + e_{interf}(t) + e_{noise}(t)\|^2}{\|e_{artif}(t)\|^2} \quad (7)$$

4. Overall heart noise or interference reduction percentage (HNRP) can assess the quality of the isolated HS [43]. The calculation formulas are shown in

$$e(\%) = \frac{E\{s_i^2(t)\} - E\{\hat{s}_i^2(t)\}}{E\{s_i^2(t)\}} \times 100\% \quad (8)$$

$$HNRP(\%) = \left(1 - \frac{|b - e|}{b} \right) \times 100\% \quad (9)$$

where b is the proportion of the HS occupied by the HLS.

3. Method

3.1. Method summary

This paper proposes an HLS separation method combining NMF and DL (see Fig. 2). The steps are as follows.

Step 1. HS extraction and classification.

- (1) Firstly, the HLS to be separated is pre-processed to transfer the HLS from the time domain to the time-frequency (T-F) domain.
- (2) Then, unsupervised blind source separation is performed on the HLS using the MCNMF decomposition and KS classification algorithms to separate the HS from it.
- (3) The separated HS are fed into the CNN to obtain the HS type, which is then used as the label information of the HLS.

Step 2. HLS separation: ECNet.

- (1) Based on the label information of the HLS (i.e. the type of HS contained in the HLS), the HLS is fed into the corresponding Sub-

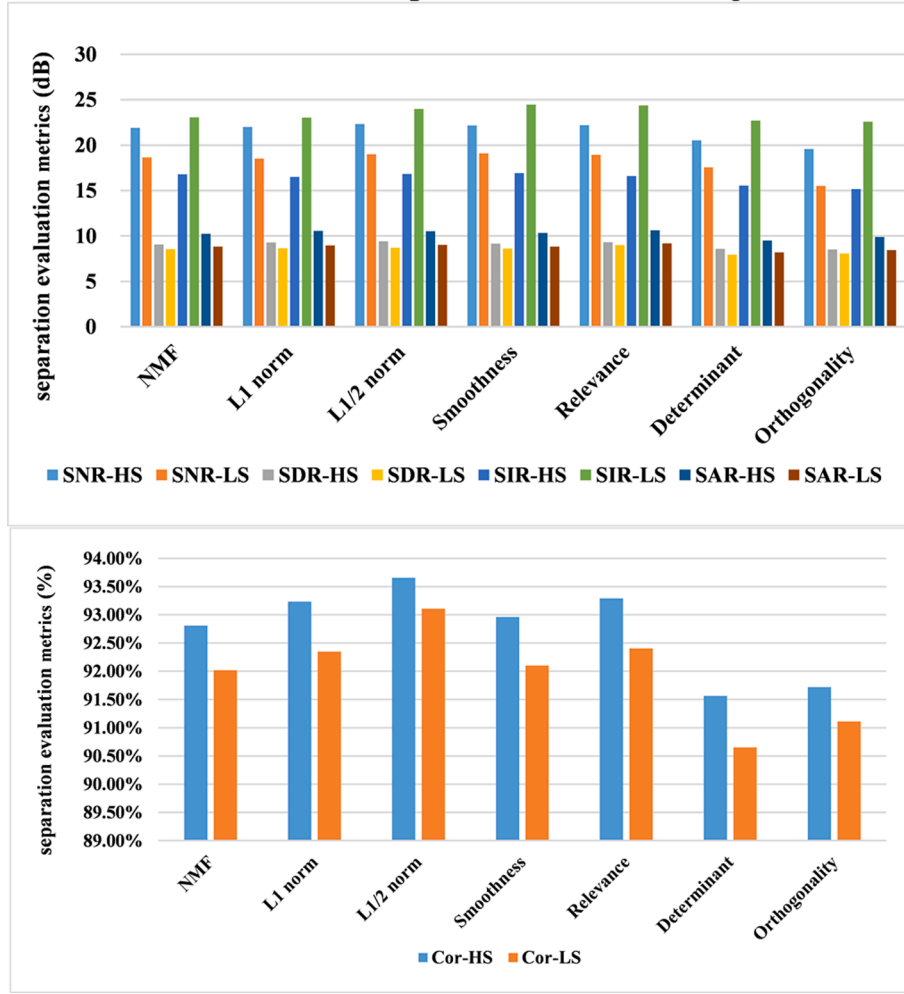


Fig. 3. The separation results based on NMF with different constraints.

neural networks (Subnet) in the ECNet, which includes several different Subnets, each trained with HLSs containing various types of HSs.

- (2) The Subnet outputs the estimated T-F mask of the HLS and reconstructs the estimated HS and the LS based on the estimated T-F mask.

3.2. HS extraction & classification

3.2.1. Pretreatment

The cardiorespiratory signal present in nature can usually be regarded as a mixed-signal $s(t)$ containing the cardiorespiratory signal, the pulmonary signal and the noise, represented by the following linear mixing model.

$$s(t) = s_c(t) + s_r(t) + n(t) \quad (10)$$

where $s_c(t)$ represents the HS, $s_r(t)$ represents the LS, and $n(t)$ means the noise. With the development of biomedical signal processing techniques and advances in electronic auscultation, ideally, the heart and lung sound signals collected by a healthcare professional through a stethoscope can be regarded as a mixed signal containing only the heart sound signal $s_c(t)$ and the lung sound signal $s_r(t)$ which can be represented by the following.

$$s(t) = s_c(t) + s_r(t) \quad (11)$$

Due to the inconsistent timing of $s(t)$ received in reality, the envelope and peak detection of $s(t)$ were extracted using cubic spline

interpolation to avoid improving the algorithm's accuracy and speed up the convergence of the neural network. Generally speaking, the first peak of the cardiorespiratory signal is the first heart sound (S1) of its cardiorespiratory component. And $s(t)$ is aligned according to this property so that S1 in all $s(t)$ lies at the same point in time. This is followed by an STFT transform of $s(t)$ to obtain the T-F spectral matrix $S(t, f)$ of $s(t)$. In subsequent articles, $S(t, f)$ is represented by Y_m .

3.2.2. MCNMF

NMF requires that the higher-order matrix V should satisfy non-negativity. V can be divided into two lower-order matrices, W and H . The modulus of Y_m is used as the NMF input.

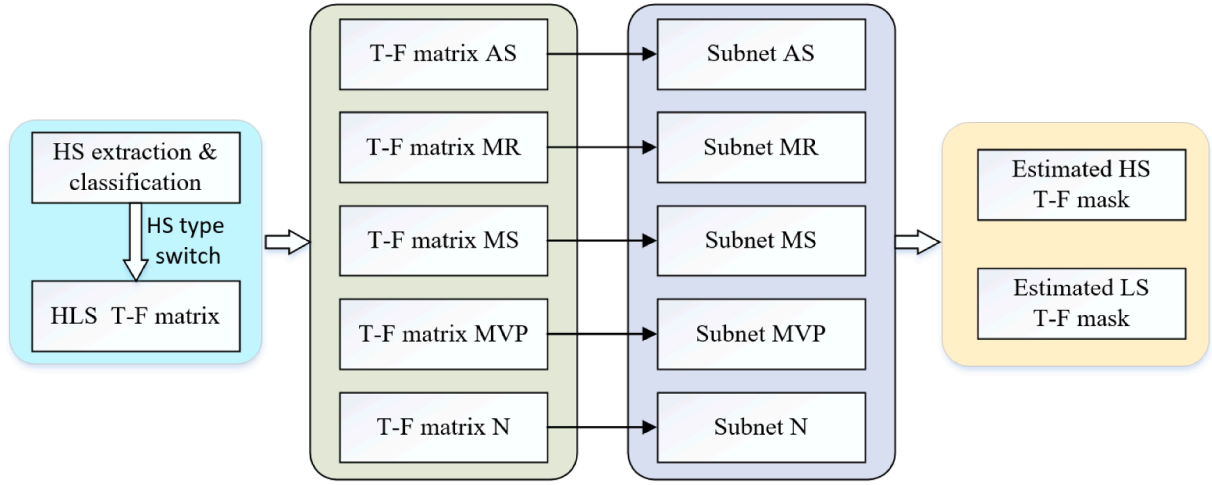
$$V = |Y_m| \approx WH \quad (12)$$

where $V \in R^{M \times N}$, $W = \{W_{ik}\} \in R^{M \times K}$, $H = \{H_{kj}\} \in R^{K \times N}$ and $1 \leq i \leq M$, $1 \leq k \leq K$, $1 \leq j \leq N$. Due to the non-negative constraints of W and H , errors occur in decomposing V . Thus, many methods have been researched to improve NMF. One of them is constrained non-negative matrix factorization (CNMF), which adds constraints to the objective function of the Euclidean distance-based NMF. The objective function is given by.

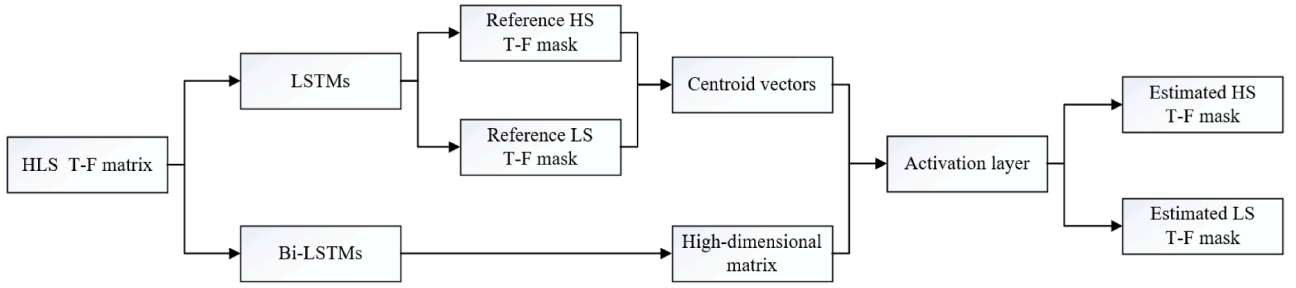
$$\min E(W, H) = \frac{1}{2} \|V - WH\|_F^2 + A = \frac{1}{2} \sum_{i,j} (V_{ij} - (WH)_{ij})^2 + A$$

$$s.t. \quad W_{ik} > 0, H_{kj} > 0 \quad (13)$$

where A is the constraint. A may act on W , H or both depending on



(a) ECNet structure diagram.



(b) Single Subnet structure diagram

Fig. 4. ECNet and its Subnet structure diagram.

Table 1

The training, validation and test dataset.

Dataset		HS					LS				HLS
		AS	MR	MS	MVP	N	AB	BB	RA	WR	
Training dataset	CNN	25	25	25	25	25	4	6	11	7	3500
	Subnet AS	25					4	6	11	7	700
	Subnet MR		25				4	6	11	7	700
	Subnet MS			25			4	6	11	7	700
	Subnet MVP				25		4	6	11	7	700
	Subnet N					25	4	6	11	7	700
Validation dataset		10	10	10	10	10	2	3	6	4	750
Test dataset		10	10	10	10	10	2	3	6	4	750

the constraint term. For HLS separation, various constraints have particular impacts on the effect. In this paper, the experiment results by adding a single different constraint to standard NMF are shown in Fig. 3. The constraints include sparse constraint based on L_1 norm [46], sparse constraint based on $L_{1/2}$ norm [47], smoothness constraint, determinant constraint [48], correlation constraint [49], orthogonality constraint, and standard NMF without any constraint [50]. Fig. 3 illustrates that $L_{1/2}$ norm, the smoothness, and the correlation constraint achieve better results than others.

We propose the MCNMF algorithm, using the best three constraints according to previous experimental results. It is based on Euclidean distance by imposing three simultaneous constraints to improve the NMF. The constraint $L(H)$ of $L_{1/2}$ norm has a more substantial sparsity effect than the L_1 norm, improving the decomposition accuracy. Furthermore, if the signal is not entirely separated, each row of the coefficient matrix H will contain other source signals, which will increase the correlation coefficient of the separated HS and LS signals.

So, we added the minimum correlation constraint $R(H)$. Finally, the

smoothness constraint $J(H)$ could increase the noise immunity of the algorithm. The objective function of the MCNMF is.

$$F(V||WH) = \frac{1}{2} \|V - WH\|_F^2 + a_1 L(H) + a_2 R(H) + a_3 J(H), \quad (14)$$

where a_1, a_2, a_3 are the regularization parameters. $L(H)$ based on $L_{1/2}$ is.

$$L(H) = \|H\|_{1/2} = \sum_{k=1}^{K,N} H_j(k)^{1/2} \quad (15)$$

where $H = \{h_1, h_2, \dots, h_K\}$ and $h_k = [h_k(n), n = 1, 2, \dots, N]$, the correlation constraint $R(H)$ is.

$$R(H) = \frac{1}{2} \left[\sum_{k=1}^K \log_{10} (HH^T)_{kk} - \log |HH^T| \right] \quad (16)$$

When $\langle h_a, h_b \rangle = 0, \forall a, b, a \neq b$, $R(H)$ is the minimum value. The smoothness constraint is.

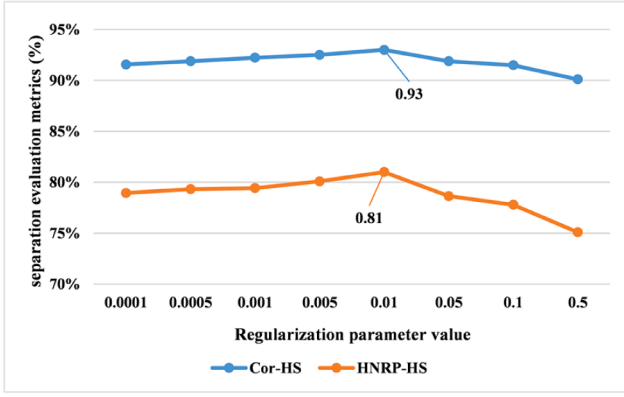
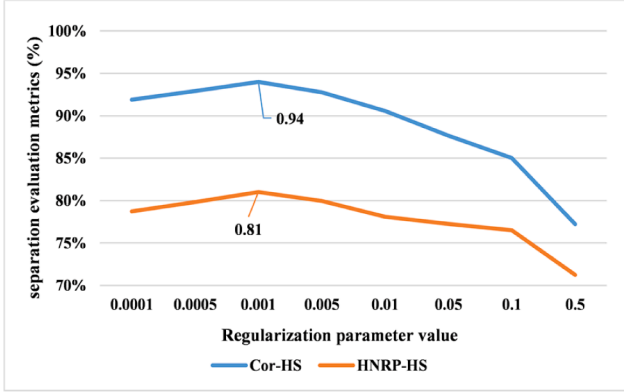
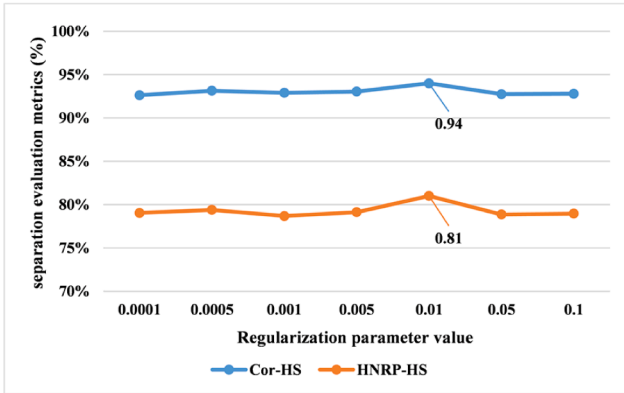
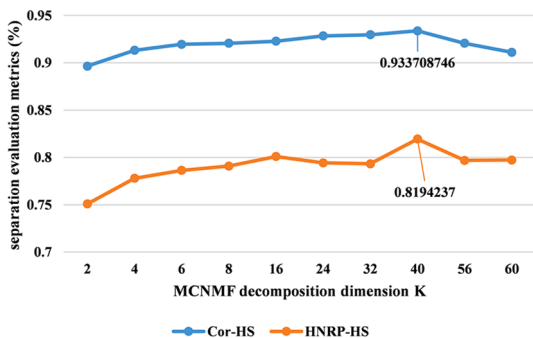
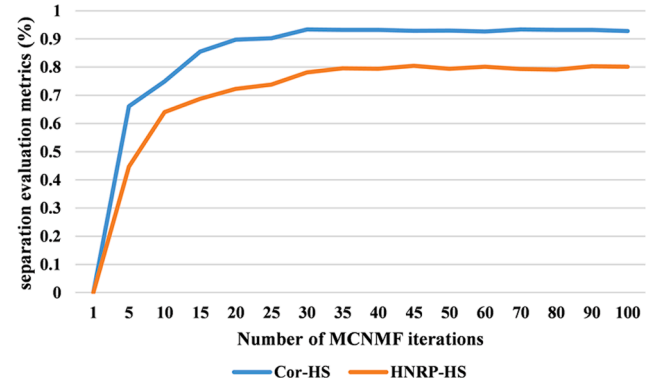
(a) The performance of the regularization parameter a_1 .(b) The performance of the regularization parameter a_2 .(c) The performance of the regularization parameter a_3 .Fig. 5. The performance of the a_1 , a_2 , a_3 in MCNMF.Fig. 6. The performance of the decomposition dimension K in MCNMF.

Fig. 7. The performance of the iterations in MCNMF.

$$J(H) = \|H\|_F^2 \quad (17)$$

In our proposed MCNMF method, the iterative formulas, Equations (20) - (26), for calculating W and H using the gradient descent method are given in Appendix A. Take them into (18) and (19), the multiplicative iterative rules of the basis matrix W and the coefficient matrix H are as shown in.

$$W_{ik} \leftarrow W_{ik} \frac{[VH^T]_{ik}}{[WHH^T]_{ik} + \varepsilon} \quad (27)$$

$$H_{kj} \leftarrow H_{kj} \frac{[W^T V]_{kj} - 2a_2 H_{kj} + a_3 [(HH^T)^{-T} H]_{kj}}{[W^T W H]_{kj} + \frac{a_1}{2} H_{kj}^{-1/2} + \varepsilon} \quad (28)$$

where ε is a small positive constant to prevent the denominator back to zero.

After the MCNMF decomposition, it is also necessary to cluster W and H . W can be regarded as a vector set consisting of K basis vectors w_k , $w_k \in R^{M \times 1}$. And H can be considered as a vector set consisting of K coefficient vectors h_k , $h_k \in R^{1 \times N}$, as shown in.

$$W = \{w_1, w_2, \dots, w_k, \dots, w_K\} \quad (29)$$

$$H = \{h_1, h_2, \dots, h_k, \dots, h_K\}^T \quad (30)$$

The product of w_k and h_k is equal to the sub-matrix X_k belongs to a certain source signal. A clustering algorithm can divide source signals belonging to the same kind together. Because the elements in X_k are non-negative, and X_k has linear additive characteristics, so the set of X_k can yield the two matrices that include HS and LS components separately. If the cluster results are right, the matrix X_c containing only HS components and the matrix X_r having LS components, inferred as.

$$X_k = w_k \times h_k, \quad k \in [1, K] \quad (31)$$

$$X_c = \sum_{k \in N_c} X_k \quad (32)$$

$$X_r = \sum_{k \in N_r} X_k \quad (33)$$

where N_c , N_r represents the number of basis vectors belonging to HS and LS, respectively.

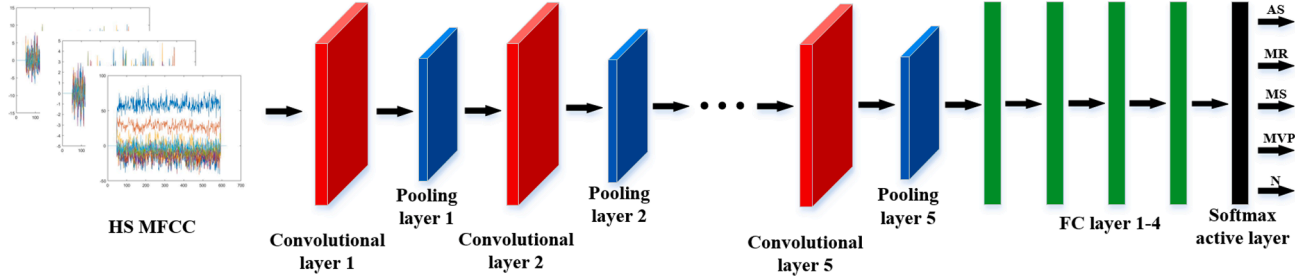
3.2.3. KS clustering algorithm

K-means clustering is one kind of clustering technique in the field of data mining for partitioning data objects. The K-means clustering algorithm determines the cluster centres randomly, and the selection of the initial values of the cluster centres can significantly affect the accuracy of the clustering. Its clustering results might be confusing when the X_c represents LS components and X_r represents HS ones. To make

Table 2

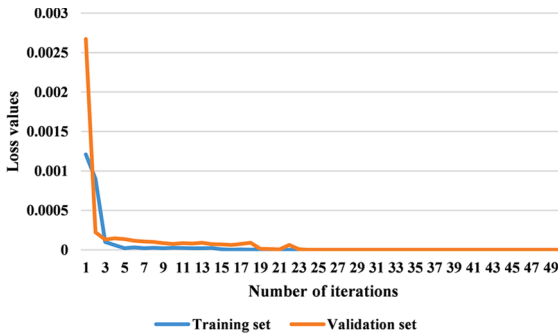
Comparison of different NMF decomposition and clustering algorithms for HLS separation.

Decomposition and clustering algorithm		SNR (dB)		Cor (%)		SDR (dB)		SIR (dB)		SAR (dB)	
		HS	LS	HS	LS	HS	LS	HS	LS	HS	LS
NMF	UC [34]	15.69	17.44	0.89	0.89	7.32	7.08	17.73	14.30	8.33	9.57
	Roll-Off [37]	16.83	18.70	0.90	0.90	7.98	7.77	19.96	14.58	8.82	9.89
	KS	21.91	18.63	0.93	0.92	9.07	8.52	16.80	23.04	10.23	8.82
SNMF [35]	UC [34]	15.69	17.52	0.89	0.89	7.29	7.16	17.64	14.32	8.34	9.51
	Roll-Off [37]	17.09	18.09	0.92	0.91	9.08	7.93	20.10	13.82	9.88	10.25
	KS	22.10	18.53	0.93	0.92	9.26	8.64	16.52	23.02	10.55	8.95
MCNMF	UC [34]	16.38	16.58	0.89	0.89	7.48	7.12	16.35	15.34	8.84	9.2
	Roll-Off [37]	17.21	18.85	0.91	0.91	8.50	7.98	19.74	13.82	9.4	10.24
	KS	22.25	18.75	0.93	0.92	9.34	8.70	16.90	23.19	10.59	9.01

**Fig. 8.** CNN network structure.**Table 3**

CNN settings.

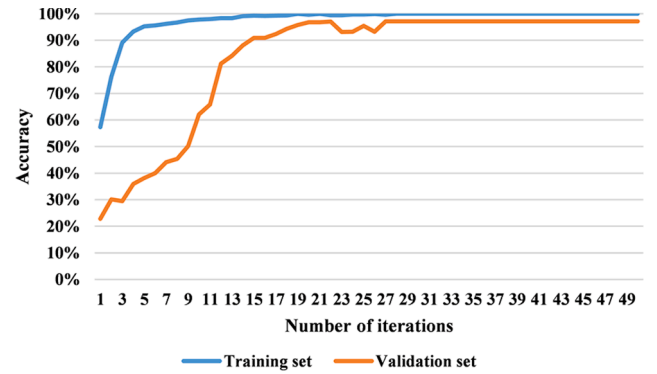
Parameters	Kernel size	Activation/Pooling	Padding	Stride	Input channels/Features	Output channels/Features
Conv. layer 1–5	(4, 2)	ReLU	1	1	3, 16, 32, 64, 128	16, 32, 64, 128, 256
Pooling layer 1–5	(4, 2)	AvgPool	1	0		
FC layer 1–4		ReLU			4608, 1024, 512, 128	16, 32, 64, 128, 256

**Fig. 9.** The loss of CNN on the training and validation dataset.

sure that X_c only contains HS and X_r only includes LS separately, our study introduces an improved clustering algorithm based on K-means and SVM, denoted as KS. The implementation steps are:

Step 1. To improve the clustering accuracy caused by the random initial clustering centers. The KS algorithm applied a 2-dimensional MCNMF decomposition of V , then calculated the clustering center w_c and w_r by the Roll-Off clustering algorithm [37]. The Roll-Off clustering algorithm can accurately fulfil the clustering due to the decomposition's low dimensionality.

Step 2. K-means clustering based on the distance of cosine similarity is performed for each w_k . For each w_k , calculate the distances to the clustering centres w_c and w_r . Categorize them as heart or lung sounds. Recalculate the clustering centres and repeat the process.

**Fig. 10.** The accuracy of CNN on the training and validation dataset.

Then, the w_k and h_k categorised as heart or lung sounds are multiplied and linearly accumulated to obtain the estimated HS T-F domain matrix X_c and the estimated LS T-F domain matrix X_r .

Step 3. For updating the clustering centres iteratively, it may result in clustering species opposite to the actual species. By SVM classifier, ensure that X_c include HS component and X_r component LS component, respectively.

3.2.4. CNN

CNN is one of the deep learning algorithms, mainly employed to exploit spatial information from the input data. To classify HS according to its different HS types in the X_c components, we make an ISTFT from X_c , calculate its MFCC, then feed it into CNN. After obtaining the type of

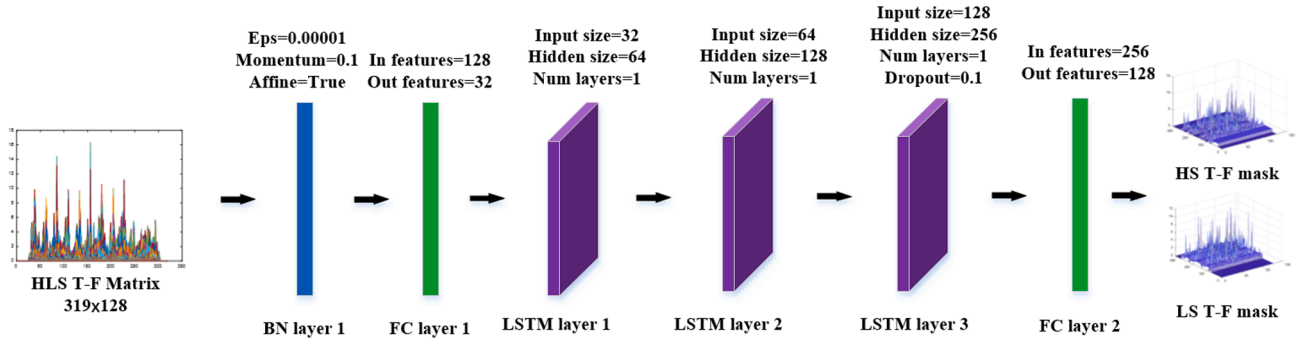


Fig. 11. LSTMs network structure.

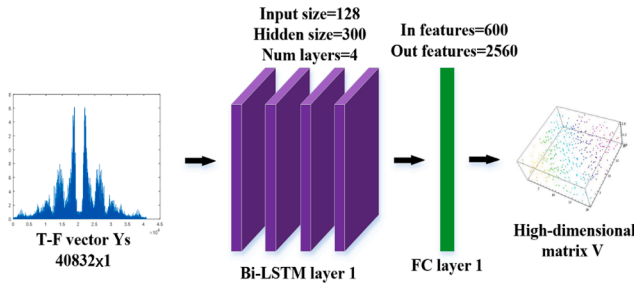


Fig. 12. Bi-LSTMs network structure.

Table 4

LSTMs settings.

Parameters	Value
Batch size	5
Optimizer	Adam
Criterion	MSE
Learning rate	0.01
LSTM Layer nonlinearity	Tanh
FC Layer 1 nonlinearity	Tanh
FC Layer 2 nonlinearity	Tanh
FC Layer 2 activation function	ReLU

Table 5

Bi-LSTMs settings.

Parameters	Value
Batch size	1
Optimizer	Adam
Criterion	MSE
Learning rate	0.001
Bi-LSTM Layer nonlinearity	Tanh
Bi-LSTM Layer bidirectional	True
FC Layer 1 nonlinearity	Tanh

HS, it is entered into ECNet along with the HLS as label information for the HLS. The structure diagram and detailed parameters of the CNN in our study are given in Section 4.1.2.

3.3. HLS separation: ECNet

The LSTM is a kind of recurrent neural network (RNN) whose unique structure can solve the problem of gradient disappearance and explosion when training long sequence signals. However, there are several types of HS in the HLS, which will show a significant variation in the time domain and frequency domain of the HLS, leading to poor separation results by the LSTM. To exploit the detailed clues in HLS while holding the advantage of the neural network, ECNet is proposed. ECNet is an

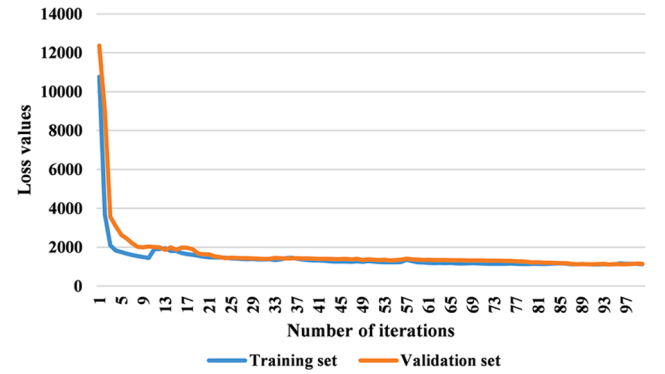


Fig. 13. The loss of LSTMs on the training and validation dataset.

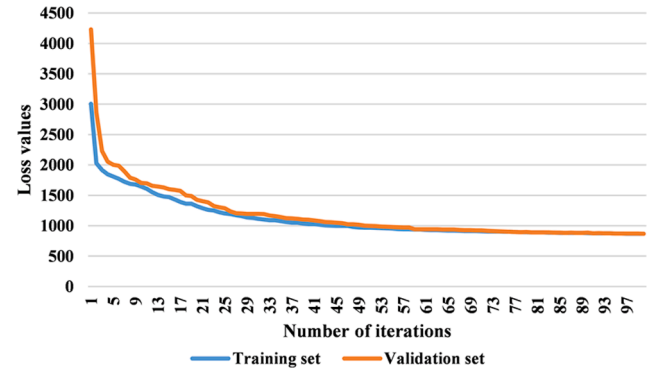


Fig. 14. The loss of Bi-LSTMs on the training and validation dataset.

LSTM-based neural network that consists of several sub-neural networks (Subnet). The structure diagram of ECNet is shown in Fig. 4. The steps of the ECNet are as follows.

Step 1. After HS extraction by MCNMF and classification by KS and CNN, according to the label information of HLS, the T-F spectrum Y_m of HLS with different HS types are switched into the related Subnet.

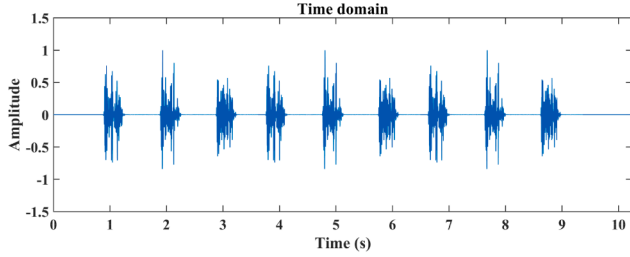
Step 2. In a single Subnet, Bi-LSTMs is one of the internal components. Its structural network with detailed parameters is given in Section 4.1.3. The T-F spectrum Y_m is transformed to a T-F vector Y_s , which is input to the Bi-LSTMs, as shown in.

$$Y_s = [x_1, x_2, \dots, x_s, \dots, x_S]^T \quad (34)$$

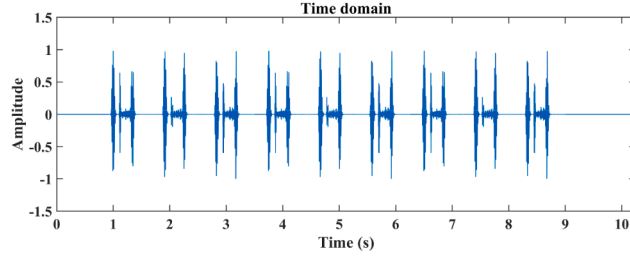
where $Y_s \in R^{S \times 1}$, x_s is the T-F point value, and S is the product of time frame T and frequency point F . Bi-LSTMs perform an embedding operation mapping each Y_s to a low-dimensional dense vector $v_k, v_k \in R^{S \times 1}$. A set of v_k consists of high-dimensional matrix V , calculated by.

Table 6
Comparison results.

Method	SNR (dB)		Cor (%)		SDR (dB)		SIR (dB)		SAR (dB)	
	HS	LS	HS	LS	HS	LS	HS	LS	HS	LS
Improved NMF [37]	16.83	18.70	90	90	7.98	7.77	19.96	14.58	8.82	9.89
NMF [34]	15.69	17.44	89	89	7.32	7.08	17.73	14.30	8.33	9.57
LSTM [40]	30.79	30.37	97	97	13.37	13.33	26.79	25.71	13.91	13.72
Proposed method	34.12	34.00	98	98	14.83	14.81	29.05	31.14	15.09	15.04

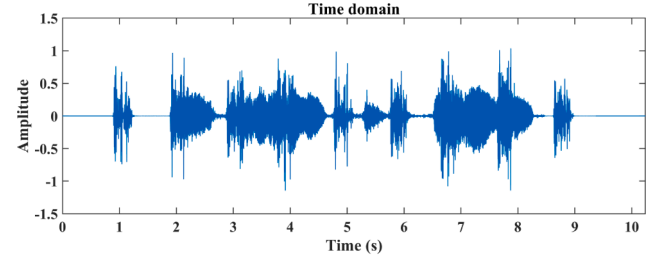


(a) Original HS and LS. Normal HS, LS type: RA.

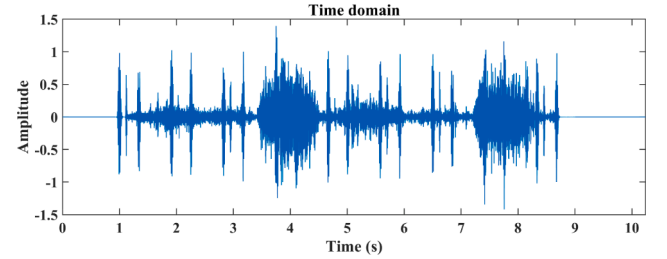


(b) Original HS and LS. HS type: MVP, LS type: BB.

Fig. 15. Original HS and LS in time domain.



(a) Mixture HLS with original HS and LS in Fig.15 (a)



(b) Mixture HLS with original HS and LS in Fig.15 (b)

Fig. 16. Mixture HLS with original HS and LS from Fig. 15.

$$V = [v_1, v_2, \dots, v_k, \dots, v_K] = \begin{bmatrix} x_1^1 & x_1^2 & \dots & x_1^K \\ x_2^1 & x_2^2 & \dots & x_2^K \\ x_3^1 & x_3^2 & \dots & x_3^K \\ \vdots & \vdots & \ddots & \vdots \\ x_S^1 & x_S^2 & \dots & x_S^K \end{bmatrix}, \quad (35)$$

$$V \in R^{S \times K},$$

$$v_k = [x_1^k, x_2^k, \dots, x_S^k]^T, 1 \leq k \leq K \quad (36)$$

where x_s^k represents the s -th T-F point in the k -th dimension space.

Step 3. Estimate the two centroid vectors standing for HS and LS in the high-dimensional matrix V . LSTMs is also one of the internal components in Subnet. Its structural network with detailed parameters is given in Section 4. Firstly, the T-F mask represents the proportion of HS or LS to HLS with the same T-F points, respectively. The T-F masks are defined as follows,

$$MASK_c = \frac{abs(Y_c)}{abs(Y_m)}, MASK_r = \frac{abs(Y_r)}{abs(Y_m)}, \quad (37)$$

where Y_c and Y_r are the T-F matrix of the original HS and LS signal $s_c(t)$ and $s_r(t)$. In the training process of ECNet, $MASK_c$, $MASK_r$, and Y_m are used to update the weight values of the LSTMs. Secondly, during the testing process of ECNet, using Y_m as input to the LSTMs, the reference T-F masks $MASK_{rc}$ and $MASK_{rr}$ of HS and LS are calculated as.

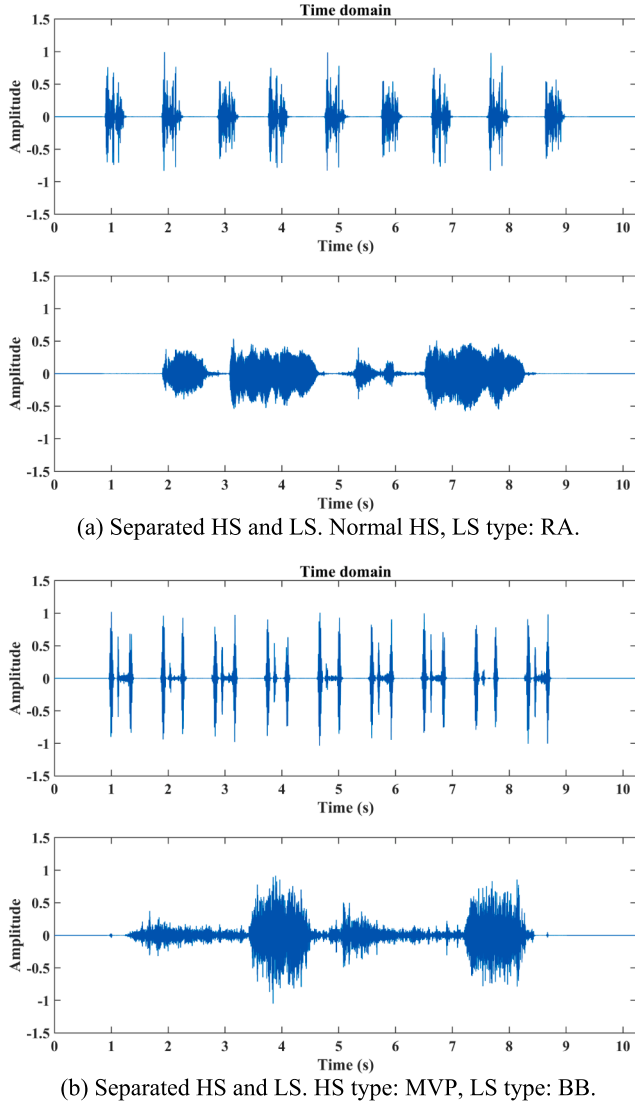


Fig. 17. Separated HS and LS in time domain.

$$MASK_{rc} = \begin{bmatrix} m_1^c \\ m_2^c \\ \vdots \\ m_S^c \end{bmatrix}, MASK_{rr} = \begin{bmatrix} m_1^r \\ m_2^r \\ \vdots \\ m_S^r \end{bmatrix}, \quad (38)$$

where $MASK_{rc} \in R^{S \times 1}$, $MASK_{rr} \in R^{S \times 1}$, m_s^c, m_s^r represents the HS and LS mask for each T-F point, $\forall m_s^c, m_s^r \in [0, 1]$. Finally, considering the different weight of the various source signals in the high-dimensional space. The two centroid vectors are derived from the $MASK_{rc}$, $MASK_{rr}$ and V as follows.

$$centroid^k = \left[\frac{x_1^k m_1^c + x_2^k m_2^c + \dots + x_S^k m_S^c}{\sum_{s=1}^S m_s^c \sum_{k=1}^K \sum_{s=1}^S (x_s^k)^2}, \frac{x_1^k m_1^r + x_2^k m_2^r + \dots + x_S^k m_S^r}{\sum_{s=1}^S m_s^r \sum_{k=1}^K \sum_{s=1}^S (x_s^k)^2} \right], \quad (39)$$

where $centroid^k \in R^{1 \times 2}$ represents the coordinates of the centroids of HS and LS in the k -th dimension.

Step 4. Use the coordinates $centroid^k$ and v_k to get $distance^k$. It indicates the distance between each point and the centroid vector in particular dimension, as shown in.

$$distance^k = v_k \cdot centroid^k \quad (40)$$

$$Distance = \sum_{k=1}^K distance^k, Distance \in R^{S \times 2}$$

Where $distance^k \in R^{S \times 2}$, $k \in [1, K]$. The magnitude of the fundamental values in the first and second columns of the $Distance$ show the similarity to the centroid of HS and LS, respectively. Activation of $Distance$ by the softmax layer gets the estimated T-F mask of HS and the estimated T-F mask of LS. The calculation formulas are as follows,

$$MASK_{ec} = \text{softmax}(Distance(:, 1)), MASK_{ec} \in R^{S \times 1} \quad (42)$$

$$MASK_{er} = \text{softmax}(Distance(:, 2)), MASK_{er} \in R^{S \times 1} \quad (43)$$

Then, the $MASK_{ec}$ and $MASK_{er}$ is expanded to two dimensions, the same size as the Y_m .

3.4. Reconstruction

The separated signal, $x_c(t)$ for HS and $x_r(t)$ for LS in time domain, can be reconstructed by ISTFT, as shown in.

$$x_c(t) = \text{ISTFT}(Y_m \odot MASK_{ec}) \quad (44)$$

$$x_r(t) = \text{ISTFT}(Y_m \odot MASK_{er}) \quad (45)$$

4. Experimental results

4.1. Experimental setup

To verify the validity of the proposed method in this work, heart-lung sound separation experiment was carried out by comparing several approaches with the same database.

4.1.1. Database

The performance of the HLS separation was evaluated on a dataset consisting of 1000 original HS signals and 193 original LS signals. The dataset of the current study is available, as shown in Appendix B. The LS used during the current study are available from the corresponding author on reasonable request.

Table 7

Performance comparison of HLS separation algorithms.

Method	SNR (dB)		Cor (%)		SDR (dB)		SIR (dB)		SAR (dB)	
	HS	LS	HS	LS	HS	LS	HS	LS	HS	LS
NMF [34]							21.13	16.78		
Improved NMF [37]					14.4		18.9		17.7	
PC-DAE [41]					8.72	7.71	14.95	13.61	10.44	10.37
Convolutional BSS [25]							30.89	21.77		
C-NMF [51]	24.11	25.06	99	99						
Proposed method	34.12	34.00	98	98	14.83	14.81	29.05	31.14	15.09	15.04

HS signals include the normal (N) and abnormal HS. Abnormal HS signals are divided into AS, MR, MS, and MVP. LS signals consist of the following types, AB, BB, WR, and RA. Triple cross-validation was employed. Table 1 shows the details of the dataset.

4.1.2. Parameters in HS extraction & classification

To use the best detection parameters of decomposition dimension K , the number of iterations and regularization parameters a_1 , a_2 , a_3 in the MCNMF, different values were tested by evaluation metrics Cor and HNR, as shown in Fig. 5 Fig. 6 Fig. 7. The results illustrate that the best parameters are $a_1 = 0.01$, $a_2 = 0.001$, $a_3 = 0.01$, $K = 40$, and iterations = 40.

Table 2 shows the results of different NMF decomposition and clustering algorithms for the HLS separation. The decomposition algorithms comprise the standard NMF, sparse constrained non-negative matrix factorization (SNMF) [35], and our proposed MCNMF. Meanwhile, the clustering algorithms include unsupervised clustering (UC) [34], Roll-Off [37], and our proposed KS. From Table 2, our proposed algorithm with MCNMF and KS clustering can achieve the best results in SNR-HS, Cor-HS, SDR-HS, SAR-HS, Cor-LS, SDR-LS, SIR-LS. It should be noted that three clustering methods exist for species in the literature [37], and only one of them is selected for comparison in this paper.

Fig. 8 shows the structure of the CNN network. Table 3 illustrates the parameter settings of the CNN. For global parameters, set batch size = 5, adam optimizer, cross entropy criterion, learning rate = 0.0003. Fig. 9 and Fig. 10 show the CNN's loss and accuracy performance. The results show that the validation accuracy is more than 95 %, proving that the CNN can classify HS according to different types effectively.

4.1.3. ECNet structure and parameters

Fig. 11 and Fig. 12 show the structure of the LSTMs and Bi-LSTMs illustrated in Fig. 4. Table 4 and Table 5 give the parameter settings of them. For the LSTMs, the batch normalization and dropout layers can speed up the training and avoid overfitting. For the Bi-LSTMs, the loss function is calculated by the $MASK_{ec}$, $MASK_{er}$, $MASK_c$, $MASK_r$. Fig. 13 and Fig. 14 show the loss values of the LSTMs and Bi-LSTMs on the training and validation dataset. They illustrate that when the number of iterations increases, the loss value tends to converge, proving that the network is effective.

4.2. Experimental results and analysis

The HLS separation method proposed in this paper was compared with three approaches: Improved NMF [37], NMF [34], and LSTM [40]. Comparison results are shown in Table 6. The experimental results show that our method outperforms other methods in all ten metrics. Our approach has the advantage of NMF's ability of unsupervised HLS separation without a priori conditions, as well as the high accuracy of LSTM for data analysis. Primarily, because there are many different HS types in the HLS, the MCNMF method and KS clustering algorithm can exploit more detailed information from HS types in HLS. Moreover, the KS clustering algorithm can distinguish the HS component from the LS component in the two clustering results. Thus, HS can be easily classified

by CNN. Finally, ECNet can achieve better HLS separation results by the previous detailed HS information, according to the proposed strategy of HLS separation followed by classification.

In the Appendix C, Fig. 15 illustrates the original HS and LS. Fig. 16 shows the mixture HLS. Fig. 17 presents the separated HS and LS results. The performances of the proposed method are then compared with previous researches on HLS separation, as shown in Table 7. The results show that our method can outperform other approaches on the evaluation metrics: SNR-HS, SNR-LS, SDR-HS, SDR-LS, SIR-LS, SAR-HS, SAR-LS.

5. Conclusion

In this paper, we propose a novel approach to heart-lung sound separation for the task of auscultation of cardiopulmonary disease. Based on NMF and deep learning techniques, our method performs an unsupervised single-channel blind source separation (BSS). The study focuses on exploiting the differences in HS of the mixture signals to achieve significant separation results. The detailed information about heart sound (HS) types in mixture sources is extracted directly by the MCNMF decomposition and the KS clustering technique. Furthermore, to recover the original sources, the CNN classifier as a switch controller inputs mixture signals with different HS types into ECNet. Compared to the state-of-the-art methods, a substantial improvement was shown in terms of SNR, SDR, SIR, SAR, and correlation coefficient. Because of the remarkable capability of our proposed method, in the subsequent work, we will expand our dataset by adding more mixture signals with other HS types, and design better neural networks to enhance the universality and scalability of the method.

CRediT authorship contribution statement

Weibo Wang: Conceptualization, Validation, Writing – review & editing, Writing – original draft, Project administration, Supervision. **Shubo Wang:** Software, Formal analysis, Investigation, Data curation, Methodology, Writing – original draft. **Dimei Qin:** Software, Formal analysis, Investigation, Data curation, Methodology. **Yu Fang:** Project administration, Conceptualization, Supervision. **Yongkang Zheng:** Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Weibo Wang reports financial support was provided by National Natural Science Foundation of China.

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Appendix A

For the objective function $F(W, H)$ of the MCNMF, the multiplicative iterative formulas for the basis matrix W and the coefficient matrix H can be derived by the gradient descent method as follows.

$$W_{ik} \leftarrow W_{ik} - \eta_W \frac{\partial E(V||WH)}{\partial W_{ik}}$$

$$H_{kj} \leftarrow H_{kj} - \eta_H \frac{\partial E(V||WH)}{\partial H_{kj}}$$

where $\frac{\partial E(V||WH)}{\partial W_{ik}}$ and $\frac{\partial E(V||WH)}{\partial H_{kj}}$ are the partial derivatives of $F(W, H)$ concerning W_{ik} and H_{kj} respectively, calculated as.

$$\frac{\partial E(V||WH)}{\partial W_{ik}} = [WHH^T - VH^T]_{ik}$$

$$\frac{\partial E(V||WH)}{\partial H_{kj}} = [W^T WH - W^T X]_{kj} + a_1 \frac{\partial L(H)}{\partial H_{kj}} + a_2 \frac{\partial R(H)}{\partial H_{kj}} + a_3 \frac{\partial J(H)}{\partial H_{kj}},$$

where the partial derivatives of the sparsity constraint term $L(H)$, the correlation constraint term $R(H)$, and the smoothness constraint term $J(H)$ concerning H_{kj} are calculated as.

$$\frac{\partial L(H)}{\partial H_{kj}} = \frac{1}{2} H_{kj}^{-1/2}$$

$$\frac{\partial R(H)}{\partial H_{kj}} = - \left[(HH^T)^{-T} H \right]_{kj}$$

$$\frac{\partial J(H)}{\partial H_{kj}} = 2H_{kj}$$

set the descent steps,

$$\eta_w = \frac{W_{ik}}{[WHH^T]_{ik}}$$

$$\eta_H = \frac{H_{kj}}{[W^T WH + \frac{a_1}{2} H^{-1/2}]_{kj}}$$

Appendix B

The heart sound data used in this article can be obtained from <https://github.com/yaseen21khan/Classification-of-Heart-Sound-Signal-Using-Multiple-Features-#matlab-code-heart-sound-signal-processing-mfcc-features>.

Appendix C

In the Appendix C, Fig. 15 illustrates the original HS and LS. Fig. 16 shows the mixture HLS. And Fig. 17 presents the separated HS and LS results.

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